



Explainable & Adaptive Credit Card Fraud Risk Intelligence System

MILESTONE PROGRESS REPORT — PHASES 0 & 1

Implementation of Credit Card Fraud Detection Using Machine Learning Algorithms

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Date
September 9, 2026

Phase 0 — Complete

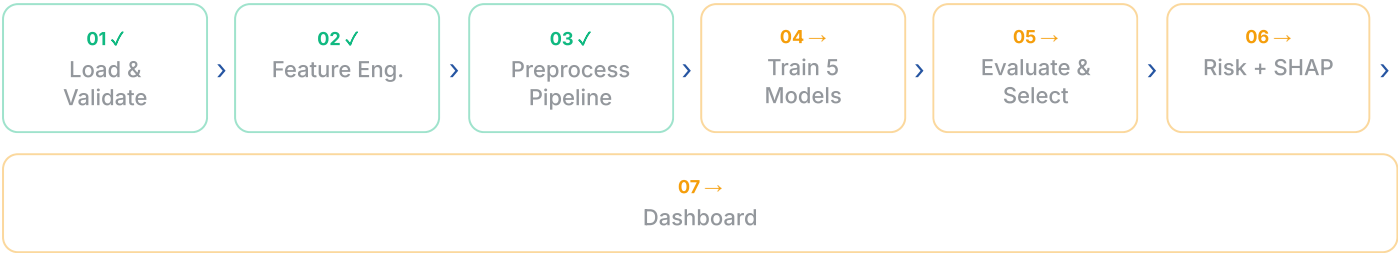
Phase 1 — Complete

Phase 2 — Next

59 / 59 Tests Passing

Architecture & Pipeline Overview

End-to-end fraud risk intelligence — from raw transaction to actionable decision



MODELS PLANNED

MODEL	IMBALANCE STRATEGY
Dummy Baseline	stratified
Logistic Regression	class_weight='balanced'
Decision Tree	class_weight='balanced'
Random Forest	class_weight='balanced'
XGBoost	scale_pos_weight

MODEL SELECTION CRITERION

Score =
0.35 × F2-score
+ 0.30 × PR-AUC
+ 0.20 × ROC-AUC
+ 0.15 × F1-score
F2-score is primary — recall-weighted, because missing a fraud is costlier than a false alarm.

RISK OUTPUT FORMAT

Score 0–30	LOW — ALLOW
Score 31–70	MEDIUM — MONITOR
Score 71–100	HIGH — REVIEW

KEY ENGINEERING PRINCIPLES

NO DATA LEAKAGE	REPRODUCIBILITY	EXPLAINABILITY
✓ Scaler fitted on training data only	✓ Random state = 42 everywhere	✓ SHAP per-transaction
✓ SMOTE applied after split	✓ Stratified splits	✓ Top-K feature contributions
✓ Threshold tuned on val set	✓ Serialized pipeline artifacts	✓ Correlation, not causation

Dataset Analysis — Verified Statistics

ULB Kaggle Credit Card Fraud Detection · All numbers obtained from actual dataset inspection

284,807	31	492	1:577
TOTAL TRANSACTIONS	COLUMNS	FRAUD TRANSACTIONS	IMBALANCE RATIO

CLASS IMBALANCE — THE CORE CHALLENGE

Legitimate — 284,315 (99.827%)	0.17%
A model predicting <i>always legitimate</i> achieves 99.8% accuracy — making accuracy a meaningless metric. This is why we use Precision, Recall, F2-score, PR-AUC instead.	

AMOUNT DISTRIBUTION

STATISTIC	VALUE
Mean	€88.35
Median	€22.00
Maximum	€25,691.16
Skewness	16.98 ← very high
Kurtosis	845.09 ← heavy tail
Zero-amount txns	1,825
Fraud mean Amount	€122.21
Legitimate mean Amount	€88.29

TOP FRAUD-CORRELATED FEATURES

V17	0.3265
V14	0.3025
V12	0.2606
V10	0.2169
V16	0.1965

DATA QUALITY

Missing values	0 — None
Duplicate rows	1,081 — Dropped
Non-numeric cols	0 — All numeric
Encoding needed	None

Feature Engineering & Preprocessing Pipeline

3 derived features, leakage-safe sklearn Pipeline, stratified 60/20/20 split

DERIVED FEATURES — RATIONALE FROM ACTUAL DATA

Log_amount	hour_of_day	is_night
log1p(Amount)	(Time % 86400) / 3600	1 if hour_of_day < 6
Amount has extreme right skew. Log transform normalises distribution and handles zero-amount transactions safely.	Dataset spans exactly 48 hours. Extracts cyclical time-of-day pattern. Fraud occurs earlier on average (80,747s vs 94,838s).	Binary flag for low-supervision window. 8.40% of all transactions occur between midnight and 6am (verified: 23,934 transactions).
Skewness: 16.98 · Kurtosis: 845.09	Range: 0.0 – 23.99 hours	Night rate verified: 8.40%

COLUMN TREATMENT

CATEGORY	COLUMNS	COUNT	TREATMENT
Scaled	Amount, Time, log_amount, hour_of_day	4	StandardScaler
Passthrough	V1 – V28	28	Already PCA-scaled
Passthrough	is_night	1	Binary, no scale
Total Features Out		33	—

SCALER STATISTICS (TRAINING DATA ONLY)

FEATURE	TRAIN MEAN	TRAIN STD
Amount	88.1606	246.4492
Time	94,935.86	47,513.34
log_amount	3.1545	1.6556
hour_of_day	14.5393	5.8411

⚠ These statistics are learned from 170,235 training rows only. Validation and test sets are transformed using these parameters — not their own statistics.

DATA SPLIT (STRATIFIED)

170,235 TRAIN ROWS 284 fraud (0.167%)	56,745 VALIDATION ROWS 94 fraud (0.166%)	56,746 TEST ROWS (HELD OUT) 95 fraud (0.167%)
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Anti-Leakage Proof & Test Results

6 leakage prevention guarantees · 59/59 automated tests passing · Inference consistency confirmed

ANTI-LEAKAGE GUARANTEES

- ✓ **Scaler fit on training only**
Pipeline.fit() called on 170,235 training rows. Val/test use the same fitted scaler.
- ✓ **Zero index overlap (train ∩ test = ∅)**
Verified at runtime: "Leakage check passed: zero index overlap"
- ✓ **Duplicates removed before split**
1,081 duplicates dropped pre-split to prevent identical rows in train and test.
- ✓ **SMOTE disabled by default**
If enabled, SMOTE is applied only to training set after split. Config-controlled.
- ✓ **Threshold tuned on validation set**
Classification threshold is optimized on val set. Test set reserved for final report only.
- ✓ **Inference consistency: max diff = 0.00e+00**
Batch transform and single-row inference produce mathematically identical output.

TEST SUITE RESULTS

59
of 59 tests passing
0 failures · 0 errors · 2.36s

TEST GROUP	TESTS
DataLoader (init, stats, columns)	11
DataValidator (hard checks, report)	13
FeatureEngineer (transform, X/y, single)	18
Pipeline (leakage, output shape)	10
Pipeline (consistency, save/load)	7
Total	59 / 59

```
===== test session starts =====
platform win32 · Python 3.13.5
collected 59 items

.....

59 passed in 2.36s
```

Deliverables & Next Steps

What was built · What was verified · What comes next

PHASE 0 — COMPLETED

- ✓ Full project scaffold (27 directories)
- ✓ All ML packages installed & verified
- ✓ Central config.yaml — single source of truth
- ✓ Logger, serialization utilities
- ✓ README, ARCHITECTURE, SETUP docs
- ✓ Dataset acquired & extracted (143 MB)

PHASE 1 — COMPLETED

- ✓ DataLoader with schema validation
- ✓ DataValidator with structured report
- ✓ FeatureEngineer — 3 derived features
- ✓ PreprocessingPipeline — leakage-safe
- ✓ Stratified 60/20/20 split
- ✓ Pipeline saved to disk (2,896 bytes)
- ✓ DATA_DICTIONARY.md with real stats
- ✓ 59 automated tests — all passing

SAVED ARTIFACTS

ARTIFACT	STATUS
preprocessing_pipeline.joblib	Saved
preprocessing_metadata.json	Saved
data/processed/train.parquet	Saved
data/processed/validation.parquet	Saved
data/processed/test.parquet	Saved
artifacts/models/ (Phase 2)	Pending

PHASE 2 — NEXT

- Train 5 models (LR, DT, RF, XGBoost, Dummy)
- Evaluate: Precision, Recall, F1, F2, ROC-AUC, PR-AUC
- Optimize threshold on validation set
- Select best model via composite score
- Final evaluation on held-out test set
- Populate MODEL_EXPERIMENTS.md with real results

VERIFIED NUMBERS AT A GLANCE

283,726	473	33	59/59
ROWS AFTER DEDUP	FRAUD IN PIPELINE	FEATURES OUT	TESTS PASSING

ACADEMIC DISCLAIMER

This system is an academic prototype developed for a semester project in Artificial Intelligence & Data Science. It is NOT suitable for real banking or financial deployment.